

ORIGINAL ARTICLE

Carotid Stent Mobility with Regard to Head Movements: In Vitro Analysis

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Given that considerable motion of the carotid artery is present during head movements, we hypothesized that a flexible stent with low torsion might be favorable to avoid stress imparted to the stent and carotid artery. Therefore, we evaluated the flexibility of different expanded carotid stents before and after deployment in a carotid artery in vitro. Subsequently, we evaluated torsion of the bare expanded stents.

Five stents (Wallstent [Boston Scientific Corp., Natick, MA], Acculink [Guidant Corp., Indianapolis, IN], Precise [Cordis Corp., Johnson & Johnson Company, Warren, NJ], Carotid SE [Medtronic AVE, Santa Rosa, CA], and Protégé [EV3, Plymouth, MN]) were tested. Flexibility was determined using a three-point bend test recording the bending load (BL) in grams required to flex the stent 25°. Increased BL implies decreased flexibility. Torsion was measured by recording the rotation load (RL) in grams required to rotate the stents 30° along its axis. Increased RL implies increased torsion.

In the bare expanded state, the median BL was 6 g (range 1–22 g). The BL increased to 38 g (range 20–41 g) after deployment in a carotid artery, with the Carotid SE (21 g) and Wallstent (36 g) showing significantly lower BL ($p < .0001$ and $p = .0016$, respectively). Overall, the RL was 11 g (range 1–76 g). Significantly higher RL was required to rotate the Wallstent (73 g) and Precise (20 g) stents ($p < .0001$).

The flexibility of the currently used stents decreases after deployment in a carotid artery irrespective of its flexibility in the bare state. Two stents showed increased torsion compared with the other stents. Limitations in both flexibility and torsion might influence the long-term performance of carotid angioplasty stenting.

Key words: *carotid, flexibility, mechanical properties, stents, torsion*

Carotid angioplasty stenting (CAS) is currently being widely embraced owing to the perceived advantages of a less invasive treatment for carotid artery occlusive disease. Several studies suggest that early complication rates are comparable to those of carotid endarterectomy.^{1,2} However, 1-year restenosis rates up to 22%, although often asymptomatic, are reported.^{1,3} The results of CAS in de novo lesions are encouraging because CAS for recurrent carotid artery occlusive disease after carotid endarterectomy seems partially responsible for the high incidence of restenosis.^{4,5}

Nevertheless, with long-term results becoming available, restenosis, even for CAS, in de novo lesions is increasing.⁶ This ongoing process of restenosis might be inflicted by stent friction during motions of the carotid artery similar to widely implicated long-term stent failure of mobile lower extremity vessels such as the external iliac and popliteal arteries.^{7–9} In a recently published study, we found average angulation changes of 21° after CAS during head movements, leading to sharp angulations up to 85° at stent-artery junctions, whereby the stented segment of the artery behaved as a stiff unit.¹⁰ This leaves the unstented segments to accommodate head movements by increased flexion, inevitably leading to friction. The same study reported

a considerable amount of torsion of common and internal carotid arteries (CCA, ICA) of 28.6° and 18.1° , respectively, while turning the head. We hypothesized that a stent design with high flexibility and low torsion, implying a low force to rotate the stent along its axis, might be favorable to avoid stress imparted to the stented and remaining unstented segments of the CCA and the ICA. The aims of this study were to compare the flexibility (or, conversely, stiffness) of commonly used carotid stents in the expanded bare state and after deployment in a porcine common carotid artery, as well as compare torsion of the stent designs within the range of motion of common head movements.

Materials and Methods

Five stents commonly used for treating carotid artery stenosis were included in this study: Wallstent (Boston Scientific Corp., Natick, MA), Precise (Cordis Corp., Johnson & Johnson Company, Warren, NJ), Acculink (Guidant Corp., Indianapolis, IN), Carotid SE (Medtronic AVE, Santa Rosa, CA), and Protégé (EV3, Plymouth, MN). All stents are self-expandable nitinol stents, with the exception of the Wallstent, which is made of elgiloy (an alloy of cobalt and chromium) and tantalum. All stents were 8 mm in unconstrained diameter and 40 mm in length. Each stent was deployed according to the manufacturer's directions. All stents were assessed for flexibility and torsion. For each stent type, two identical stents were tested, and measurements were made in triplicate. All measurements were carried out at a temperature of 37°C to ensure that the nitinol stents were within their optimal operating conditions.

Flexibility

Stents were bent up to an angulation of 25° , representing the average angulation change in the carotid artery during head movements using the three-point bend

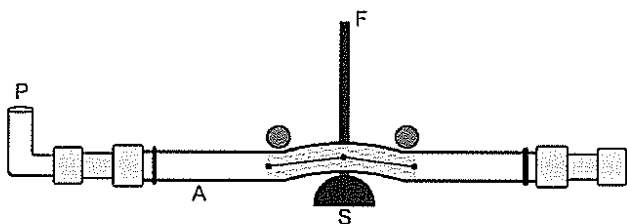


Figure 1. The modified three-point bend test for measuring stent flexibility. Force (F) is applied to an element with a convex surface (S), resulting in a 25° angulation of the stent. The artery (A) was placed into the testing device, and a 0.9% saline drip with a pressure-bag (P) was used to simulate an in vivo blood pressure of 80 mm Hg.

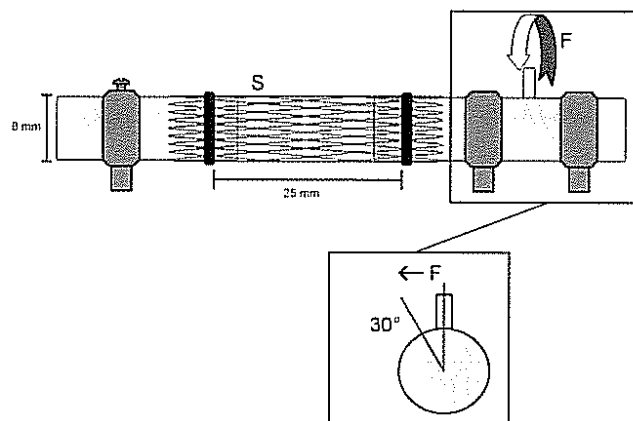


Figure 2. The testing device for measuring torsion. Force (F) is applied to rotate the stent (S) 30° along its axis.

test method previously described by Ormiston and colleagues.^{10,11} In our study, we adjusted the setup by applying a more diffuse physiologic traction using an element with a convex surface instead of the "hook" representing the third point in the Ormiston and colleagues model (Figure 1). The bending load (BL) in grams required to flex each stent to 25° was determined. Increased BL implies decreased flexibility. Healthy CCAs were harvested from 10 pigs that were used for another study (median weight 90 kg, range 88–95 kg; median age 6 months); the tests were performed within 2 hours after obtaining the arteries. The artery itself was cut into a standardized length of 100 mm and was placed into the testing device. The internal diameters of the arteries were 7 mm (range 6–7.5 mm). A 0.9% saline drip with a pressure-bag was used to simulate an in vivo blood pressure of 80 mm Hg. First, flexibility of the carotid artery itself was evaluated to avoid variances between the carotid arteries, interfering with our results. Subsequently, in each carotid artery, one stent was deployed by the same investigator. Flexibility measurements were performed at the center of the stent. Thereafter, the artery was sacrificed; the expanded stent was placed in the testing device, and flexibility measurements of the bare stents were done.

Torsion

Bare expanded stents were placed into a testing device constructed for torsion measurements (Figure 2). The stent was fixated at two points, 25 mm apart. The rotation load (RL) in grams required to rotate the stent 30° was measured. Increased RL implies increased torsion.

In both tests, an unster was used with an accuracy of ± 2 g of the absolute value and a range of

0 to 100 g. Calibration was performed using standard weights (2, 10, 20, 50, 75, and 100 g; Netherlands Calibration Organization/International Standards Organization certified).

Statistical Analysis

The results were expressed as the median (range) of the triplicate measurements of the two identical stents. To analyze differences in the flexibility of the carotid arteries, we compared the repetitive measurements of the carotid arteries by means of analysis of variance (ANOVA), with the individual carotid arteries as the between factor and the repetitions as the within factor. The differences between the several types of bare stents and the stents after deployment in carotid arteries were evaluated using one-way ANOVA with the factor type of stent with post hoc Bonferroni tests. Statistical analyses were performed with SPSS, version 11.0 (SPSS, Chicago, IL). A p value $< .05$ was considered significant.

Results

In the bare expanded state, the median BL was 6 g, with a wide spread varying from 4 g (range 1–5 g, Wallstent) to 20 g (range 18–22 g, Acculink), depending on the stent design. The BLs for the Acculink and Protégé stents were significantly higher (20 g and 14 g, respectively) in the bare state compared with the other stents ($p < .0001$). After deployment in the carotid artery, the overall BL increased to 38 g (range 20–41 g). The BL for the Carotid SE and Wallstent stents was significantly lower, 21 g (range 20–22 g; $p < .0001$) and 36 g (range 34–37 g; $p = .0016$) compared with the other stents. The BLs of bare

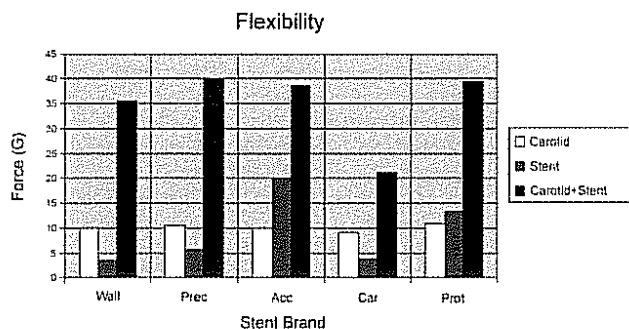


Figure 3. Comparison of flexibility of the carotid artery: the five bare expanded stent designs and subsequent flexibility of the stents while deployed in the carotid artery. Flexibility is expressed as bending force in grams required to produce 25° of flexion. Values are expressed as the median. Accu = Acculink; Caro = Carotid SE; Prec = Precise; Prot = Protégé; Wall = Wallstent.

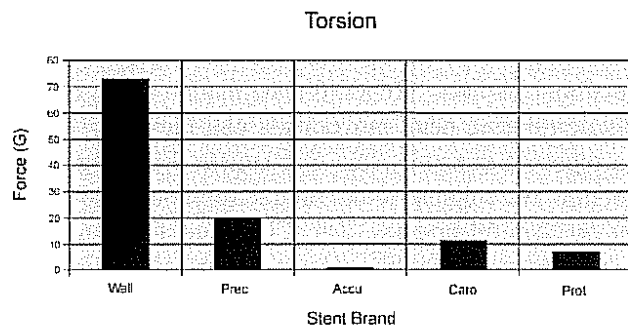


Figure 4. Comparison of torsion of the five stent designs. Torsion is expressed as rotation force (g) required to rotate the stent 30° along its axis. Values are expressed as the median. Accu = Acculink; Caro = Carotid SE; Prec = Precise; Prot = Protégé; Wall = Wallstent.

carotid stents were not significantly related to the BLs after deployment in the carotid arteries ($p = .089$). No significant difference was found between the flexibility of the carotid arteries itself in all five groups ($p = .14$). A summary of all of the bending tests is shown in Figure 3.

With regard to the torsion tests, all stents were capable of rotating along their axis, 11 g (range 1–76 g), although a significantly increased RL was applied on the Wallstent, 73 g (range 65–76 g), and the Precise, 20 g (range 19–22 g), to achieve a rotation of 30° ($p < .0001$) (Figure 4). None of the stents showed signs of deforming or collapse during the flexibility or torsion test.

Discussion

In a previous magnetic resonance angiography study describing the mobility patterns of the carotid artery after CAS during head movements, we demonstrated a complete lack of flexibility of the stented segment, although the carotid artery itself was highly flexible.¹⁰ Angulations at stent junctions aggravated by bending the head and increased torsion in the unstented segment of the ICA were observed after rotation of the head.¹⁰

Angulations and increased torsion might cause reduced patency because disturbed flow shear rates and a decreased rate of endothelialization by irregularities at the junction between the stent surface and the inner lumen of the vessel wall result in increased intimal formation.^{12,13} The extent of intimal hyperplasia has been shown to correlate with the degree of injury by penetration of the stent struts into the layers of the vessel wall.¹⁴ Obviously, when the stented segment is incapable of torsion, the stent edges

implanted in healthy landing zones of the CCA and ICA might cause deep injury to the arterial wall by torsional motions. Repetitive angulation changes and friction during head movements will inevitably result in repeated injury by this mechanism. With regard to torsion, we emphasize the unique rotational capacities of the carotid arteries. The CCA and ICA are subjected to considerable torsion shear (approximately 28° and 16°) by rotation of the head during average head movements.¹⁰

To avoid such an injury, comparable flexibility and torsion of the stented and unstented segments of the carotid artery might be the ideal option. Caution in this matter is required because experimental results conflict. Studies by Palmaz and Barth and colleagues stated that the parameter of flexibility might hamper the development of a thin neointima.^{15,16} The pulsatile nature of a flexible stent may induce platelet sloughing, embolization, and reaccumulation, thus leading to an irregularly thick subsequent platelet substrate and, eventually, a thick neointima. Another study showed the contrary by comparing the same stent design with differing longitudinal flexibility in swine iliac arteries showing less neointimal buildup in the flexible stent after 5 weeks.¹⁷

In the case of a mobile artery such as the carotid, some flexibility is, in our opinion, preferable because sharp angulations with disturbance of the luminal geometry and repetitive trauma at stent edges are to be avoided. Several studies have assessed the mechanical properties of bare expanded stents.^{11,18,19} In those, the flexibility and torsion of bare stents were objectified, and excellent results were obtained. However, in our study, increased stiffness was observed after stents were deployed in carotid artery specimens irrespective of their excellent flexible capacities in the bare state. The potential advantages especially provided by the flexible Wallstent, Precise, and Protégé stents decreased after the stents were inserted into a carotid artery. These variations are inevitably related to the gauge (thickness of wire), type of wire, and surface area of the stent. A similar finding of decreased compliance and progressive stiffness is observed after deployment in canine iliac arteries.²⁰

A limitation of this bench-test study is the lack of comparability between the porcine and the human carotid artery, whereby, in the latter, flexibility and torsion are likely to be decreased owing to the presence of atherosclerotic plaque. However, as stent placement extends into both the proximal and distal undiseased parts of the carotid artery, which represent the areas of maximal shear stress in our study, our

findings are likely to bear some resemblance to the human situation.

In conclusion, despite the highly flexible capacities of currently used stents *ex vivo*, decreased flexibility is present after deployment in the porcine carotid artery. Moreover, some of the currently used stents displayed increased torsion compared with the other stents. Restrictions in flexibility and torsion might inevitably lead to frictional forces and repetitive trauma, especially at the stent-vessel junctions, potentially causing intimal hyperplasia, stent migration, and stent breakdown. Although the restenosis rates for CAS in *de novo* lesions are encouraging, stent interaction with the mobile carotid artery can result in disappointing restenosis rates in the long term. For now, use of short stents limited to the diseased vessel wall might be preferable. Effort should be directed toward development of carotid stents with high flexibility and low torsion *in vivo*.

References

1. Brown MM, Rogers J, Bland JM, CAVATAS Investigators. Endovascular versus surgical treatment in patients with carotid stenosis in the Carotid and Vertebral Artery Transluminal Angioplasty study (CAVATAS): a randomised trial. *Lancet* 2001;357:1729–37.
2. Criado FJ, Lingelbach JM, Ledesma DF, Lucas PR. Carotid artery stenting in a vascular surgery practice. *J Vasc Surg* 2002;35:430–4.
3. Christiaans MH, Ernst JMPG, Suttorp MJ, et al. Restenosis after carotid angioplasty and stenting: a follow-up study with duplex ultrasonography. *Eur J Vasc Endovasc Surg* 2003;26:141–4.
4. Leger AR, Neale M, Harris JP. Poor durability of carotid angioplasty and stenting for treatment of recurrent artery stenosis after carotid endarterectomy: an institutional experience. *J Vasc Surg* 2001;33:1008–14.
5. Setacci C, Pula G, Baldi I, et al. Determinants of in-stent restenosis after carotid angioplasty: a case control study. *J Endovasc Ther* 2003;10:1031–8.
6. Lal BK, Hobson RW II, Goldstein J, et al. In-stent recurrent stenosis after carotid artery stenting: life table analysis and clinical relevance. *J Vasc Surg* 2003;38:1162–8.
7. Cikrit DF, Dalsing MC. Lower-extremity arterial endovascular stenting. *Surg Clin North Am* 1998;78:617–29.
8. Powell RJ, Fillinger M, Bettmann M, et al. The durability of endovascular treatment of multisegment iliac occlusive disease. *J Vasc Surg* 2000;31:1178–84.
9. Ramaswami G, Marin ML. Stent grafts in occlusive arterial disease. *Surg Clin North Am* 1999;79:597–609.
10. Vos AW, Linsen MA, Marcus JT, et al. Carotid artery dynamics during head movements: a reason for concern with regard to carotid stenting? *J Endovasc Ther* 2003;10:862–9.
11. Ormiston JA, Dixon SR, Webster MW, et al. Stent longitudinal flexibility: a comparison of 13 stent designs before and after balloon expansion. *Catheter Cardiovasc Interv* 2000;50:120–4.
12. Carlier SG, van Damme LC, Blommerde CP, et al. Augmentation of wall shear stress inhibits neointimal hyperplasia after stent implantation: inhibition through reduction of inflammation? *Circulation* 2003;107:2741–6.
13. Hamuro M, Palmaz JC, Sprague EA, et al. Influence of stent

- edge angle on endothelialization in an in vitro model. *J Vasc Interv Radiol* 2001;12:607–11.
14. Schwartz RS, Huber KC, Murphy JG, et al. Restenosis and the proportional neointimal response to coronary artery injury: results in a porcine model. *J Am Coll Cardiol* 1992;19: 267–74.
 15. Palmaz JC. Intravascular stents: tissue-stent interactions and design considerations. *AJR Am J Roentgenol* 1993;160:613–8.
 16. Barth KH, Virmani R, Froelich J, et al. Paired comparison of vascular wall reactions to Palmaz stents, Strecker tantalum stents, and Wallstents in canine iliac and femoral arteries. *Circulation* 1996;93:2161–9.
 17. Fontaine AB, Spigos DG, Eaton G, et al. Stent-induced intimal hyperplasia: are there fundamental differences between flexible and rigid stent designs? *J Vasc Interv Radiol* 1994;5: 739–44.
 18. Duda SH, Wiskirchen J, Tepe G, et al. Physical properties of endovascular stents: an experimental comparison. *J Vasc Interv Radiol* 2000;11:645–54.
 19. Dyet JF, Watts WG, Ettles DF, Nicholson AA. Mechanical properties of metallic stents: how do these properties influence the choice of stent for specific lesions? *Cardiovasc Interv Radiol* 2000;23:47–54.
 20. Back M. Changes in arterial wall compliance after endovascular stenting. *J Vasc Surg* 1994;19:905–11.